



# RaktSanket

*AI-Based Predictive Blood Availability and Donor Response System*

Seminar Report | Seventh Semester, B.Tech. (Artificial Intelligence and Data Science)

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# Introduction

Blood cannot be manufactured — it only comes from people who choose to donate. Every unit used in surgery, an accident, or childbirth begins as a voluntary donation, so hospitals depend entirely on donors being available and reachable at the right time.

India needs about 14–15 million units of blood every year, but only around 11–12 million units are collected. This gap becomes most serious during festivals, exam season, and the monsoon — yet current systems usually notice a shortage only after it has already happened.

## India's Yearly Blood Gap

**2.6M**

units short every year

**15M**

needed / year

**11M**

actually collected

*Source: Government blood-donation data*

# The Problem

*Why today's blood-donation systems fall short*



## No Early Warning

Shortages are noticed only after stock has already run low.



## Same Message to Everyone

Every donor gets the same alert, whether they are likely to respond or not.



## No Explanation

Donors and staff are not told why an alert was raised or a donor was picked.



## Donor Fatigue

The same helpful donors are messaged again and again until they stop responding.

### How Alerts Work Today

1

Stock drops below the safety level

2

Staff notices the drop after the fact

3

One message is sent to the entire donor list

4

No ranking, no explanation, no fatigue check

# The Solution

RaktSanket predicts a shortage before it happens and tells staff exactly which donors to contact, in what order, and why.

## CBSRI

*Composite Blood Shortage-Response Index*

*One simple weekly score, for every district and blood group*

1

### Shortage Forecast

Will this area run short soon?

2

### Donor Response

How likely is each donor to say yes?

3

### Fatigue Check

Has this donor been asked too often?

# Motivation

*Why this project matters*

1

## Shortages Are Predictable

Gaps often line up with festivals, exams, and the monsoon — patterns a system can learn and plan for.

2

## Donors Get Tired

Contacting the same willing donors again and again wears them out and wastes their time.

3

## Trust Needs Explanation

A plain-language reason for every alert helps staff and donors trust and act on it.

4

## Every Unit Saves a Life

A shortage that is caught early can mean the difference between a delay and a preventable death.

# Objectives

**01**

Build the CBSRI score for every district and blood group

**02**

Forecast shortages 7–14 days ahead of time

**03**

Predict how likely each donor is to respond

**04**

Explain every prediction in plain, simple language

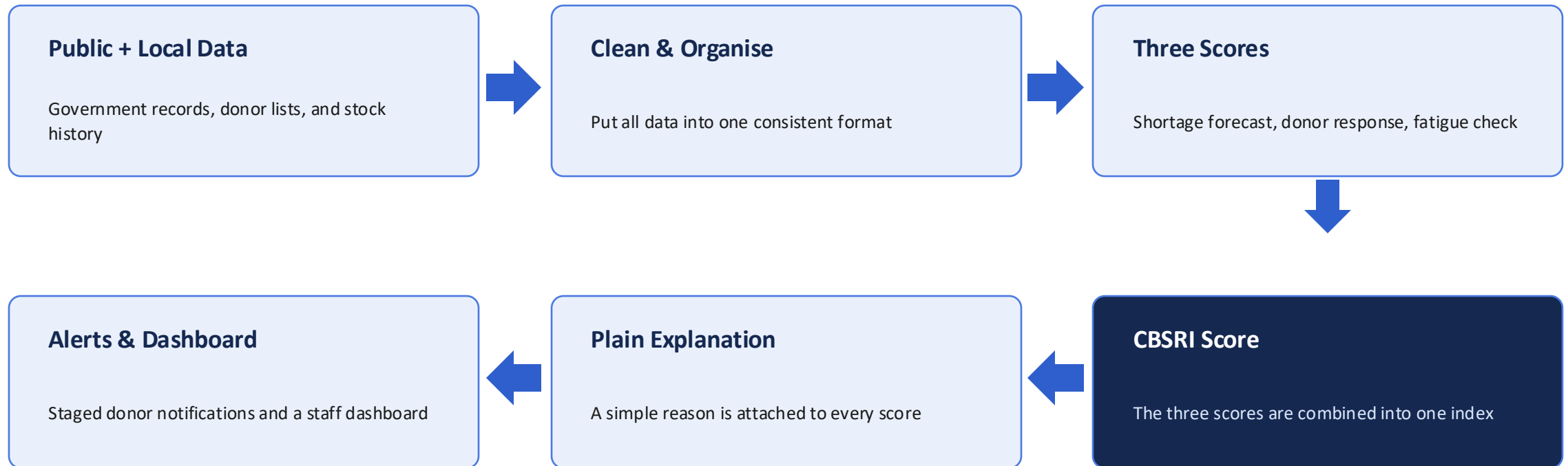
**05**

Send donor notifications in careful, staged groups

*Five objectives, working together as one connected system.*

# System Architecture

*How data flows through RaktSanket, step by step*



# Technologies and Algorithms

*The tools RaktSanket is built with, explained simply*

1

## Python & JavaScript

The programming languages used to build the system

2

## Time-Series Forecasting

Looks at past patterns to predict shortages ahead of time

3

## XGBoost

Learns from donor history to predict who is likely to respond

4

## SHAP

Turns a prediction into a plain-language reason

5

## React & FastAPI

Power the on-screen dashboard and behind-the-scenes logic

6

## PostgreSQL

Safely stores donor and stock records

# Key Features

## Predicts Early

Flags a possible shortage 7–14 days before it happens, not after.

## Ranks Donors Fairly

Reaches out first to the donors most likely to respond.

## Explains Itself

Every alert and every donor rank comes with a plain-language reason.

## Protects Tired Donors

Donors contacted too often are given a lower priority for a while.

## Sends Alerts in Stages

Notifies a small group first, and only expands if needed.

## Works on a Simple Budget

Built with free, open data and tools any district can afford.

# Conclusion

RaktSanket brings together shortage prediction, donor ranking, plain-language explanations, and staged notifications into one connected system — something existing platforms like e-RaktKosh do not offer today.

## Future Scope

- Testing against real past shortage events
- Alerts and explanations in regional languages
- ID-based donor verification
- A pilot run with a real district blood bank

*The core idea is simple:*

*predict early,*

*contact the right donor,*

*and explain why —*

*so fewer patients are ever left  
waiting for blood.*



# Thank You

*RaktSanket — AI-Based Predictive Blood Availability and Donor Response System*

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